Mado · Director's Viewfinder

User manual

Version 0.2.32 beta · November 2026

1. What Mado is

Mado (窓, "window" in Japanese) turns your phone camera into a **director's viewfinder**. You pick a camera, a recording mode and a lens, and Mado draws over the live image the frame that camera would see. It is for composing shots on a *location scout* without carrying gear, and for writing down everything another person needs to repeat the shot with the real camera.

Mado does not record video: it takes **photos of the frame** and adds the technical data to them.

> **Important.** The frame is a simulation. Its accuracy depends on the calibration of your phone's lens (see chapter 10). Where a lens is not calibrated, Mado says so.

2. Before you start

- **Phone:** Android 10 or later.
- **Permissions:** camera (required) and location (optional, only to record the place and the heading). Mado has **no internet permission**: nothing leaves your phone.
- **Calibrated phones in this version:** Samsung Galaxy S26 Ultra (main lens and both telephoto lenses; the ultra-wide is still to be measured) and Sony Xperia 1 III (SO-51B, all three rear lenses). On any other phone Mado uses the data the phone reports and tells you so.
- **Installing the beta:** you receive the APK file from whoever invited you. Open it and allow installing from unknown sources. Install it over the previous version, without uninstalling.

3. First steps (5 minutes)

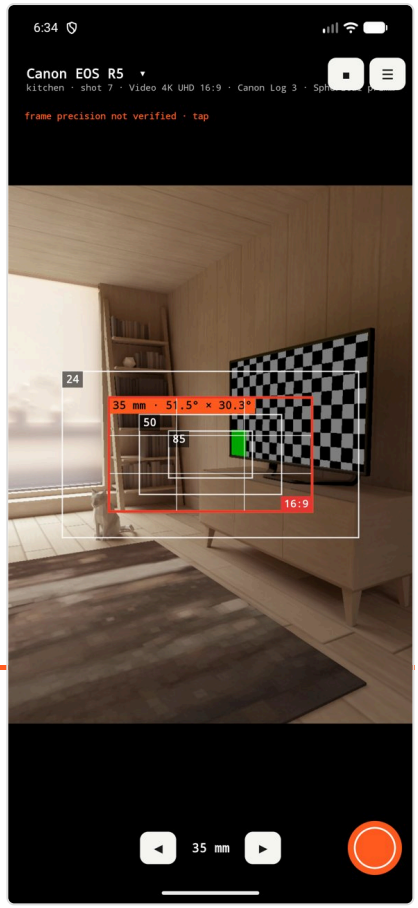
1. Open Mado. It starts on the **Canon EOS R5**.
2. Tap **start preview** (the orange button in the middle).
3. Look at the frames drawn over the image: the **orange** one is the current focal length and the **white** ones are the rest of your focal kit.
4. Change focal length with the ◄ ► arrows at the bottom.
5. Tap the **shutter** (the orange circle). The photo is saved in `Pictures/Mado`.

4. The viewfinder screen

- **Top left, the camera.** It shows the camera name and below it the scene and shot number, the recording mode, the colour profile and the lens set. **Tap it to open the options panel.**
- **Top right:** ■ stops the preview and ≡ opens the menu.
- **The frames:** each one is a focal length of your kit; the orange one is the current one. If the camera you simulate sees **wider** than your phone, the frame fills the viewfinder and the phone image shrinks

(you will see the padding).

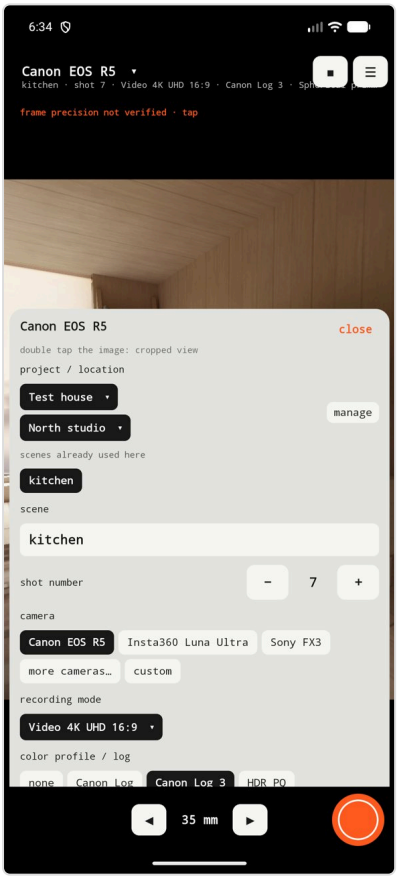
- **Bottom:** ◄ focal length ► and the **shutter**. With the phone in landscape the keys move to the side.
- **Tapping the image** focuses and meters exposure at that point. A marker shows the distance the phone focused at (for example "1.33 m"). The focus stays locked until your next tap.
- **Double tap:** cropped view, which shows the image **as it would come out** in the current frame. Tap the orange notice to go back.
- **The notice "frame precision not verified · tap"** appears when the phone lens is not calibrated (chapter 10).



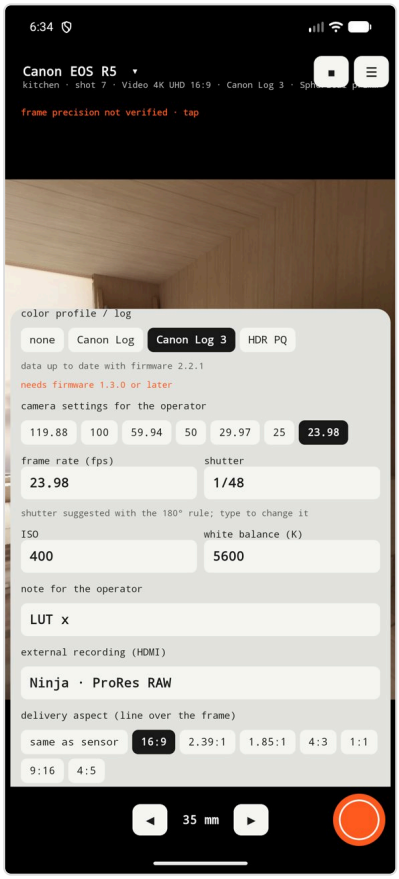
The viewfinder with the preview running

5. Choosing camera, mode and lens (options panel)

Tap the camera name, top left. The panel scrolls up and down.



The options panel, top



Further down the panel

5.1 Camera

The three main cameras (**Canon EOS R5**, **Insta360 Luna Ultra** and **Sony FX3**) are in sight; the rest are under **more cameras…**. With **custom** you type the sensor width and height in millimetres.

5.2 Recording mode

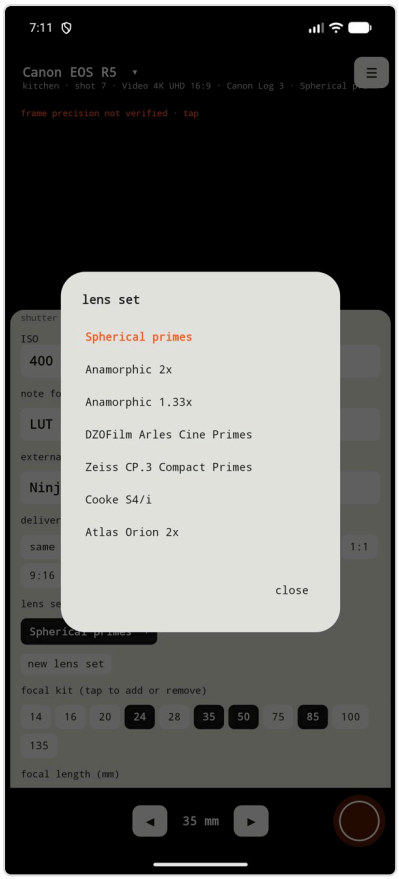
The field of view depends on the **sensor area each mode uses**, not only on the camera: recording 4K across the full sensor width is not the same as with a crop. Each camera comes with its modes and the official source. Where the maker does not publish the exact area, Mado marks it as **derived** (calculated by us). The information follows the newest firmware version we know of, and the panel says, for example, "data up to date with firmware 2.2.1". If a format needs a minimum firmware, it tells you.

5.3 Colour profile / log and Picture Profile

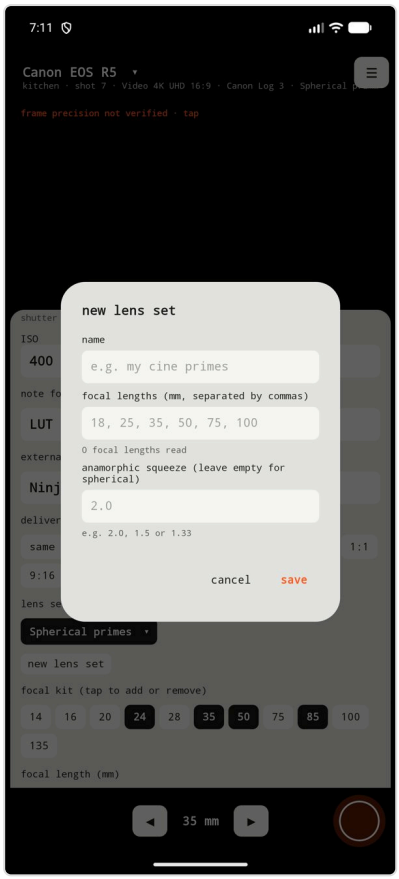
Choose the profile you will record (Canon Log, Canon Log 3, S-Log3, I-Log, HDR PQ…). It appears next to the mode, in the viewfinder and in the photo bar. On the **Sony FX3** you can also pick the **Picture Profile** (PP1 to PP6, PP10 and PP11); what each PP contains is set by whoever operates the camera, so describe it in the note (chapter 6).

5.4 Lens set and focal kit

- **Lens set:** pick one from the list. There are generic ones (spherical and anamorphic 2x and 1.33x) and sets with a **real name**: **DZOFilm Arles**, **Zeiss CP.3**, **Cooke S4/i** and **Atlas Orion 2x**, each with the source of its focal lengths. With anamorphic, the frame widens by the squeeze factor.
- **Your own sets:** with **new lens set** you type the name, the focal lengths separated by commas and, if it is anamorphic, its squeeze. They appear at the end of the list with ★ and are corrected or deleted with **edit this set**.
- When you choose a set, the drawn kit becomes the focal lengths closest to 24, 35, 50 and 85 mm. The focal lists of the named sets come from retailer listings and press, not always from the maker: check the set you carry and, if it differs, create your own.

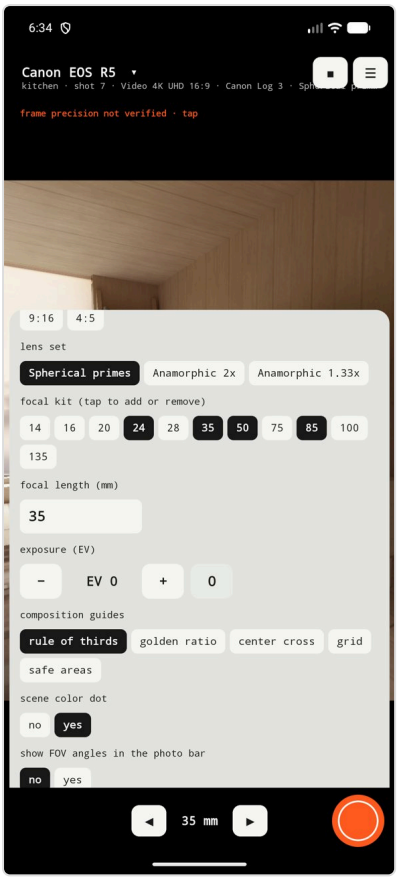


The lens set list

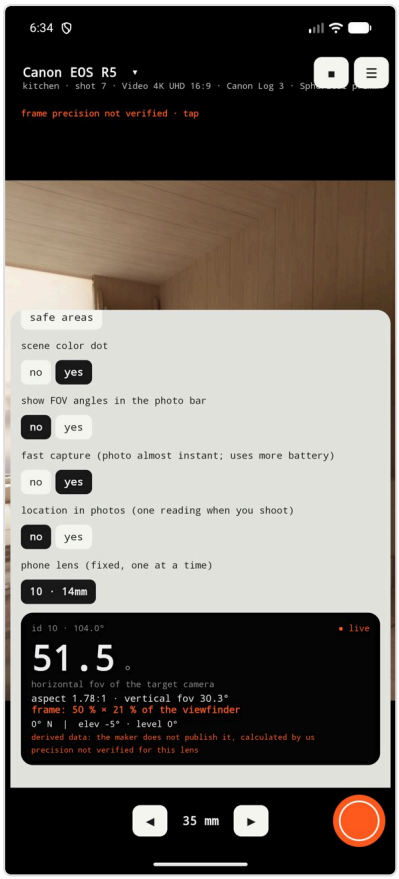


Creating your own set

- **Focal kit:** tap the numbers to add or remove focal lengths; all of them are drawn at once.
- **Focal length (mm):** the current focal length. On fixed-lens cameras (such as the Luna Ultra) you choose the built-in lens and the focal length is the equivalent.



More panel options



Exposure, guides and location

5.5 Delivery aspect

Draws a **red line** with the final delivery format (16:9, 2.39:1, 1.85:1, 4:3, 1:1, 9:16 or 4:5) over the sensor. For example, 16:9 over a 3:2 sensor.

5.6 Exposure, guides and other settings

- **Exposure (EV):** compensates the phone's exposure. It applies to the preview and the photo.
- **Composition guides:** rule of thirds, golden ratio, centre cross, grid and safe areas.
- **Scene colour dot:** a different colour dot for each scene on the photo's tag (can be turned off).
- **Show FOV angles in the photo bar:** off by default.
- **Fast capture:** the photo is taken almost instantly (uses more battery).
- **Location in photos:** one GPS reading when you shoot.
- **Phone lens:** the phone lens used as the base. Mado starts from the calibrated lens.

6. Settings for the operator

The person doing the scout is not always the one operating the camera. This section stores what the operator needs:

- **Frame rate (fps):** pick one of those the mode offers, or type it.

- **Shutter:** suggested automatically with the **180° rule** (1/48 for 24 fps, 1/60 for 30 fps, 1/50 for 25 fps). Type another value and yours stays.
- **ISO** and **white balance (K).**
- **Picture Profile** (on the FX3).
- **Note for the operator:** for example the LUT or the ND filter.
- **External recording (HDMI):** the recorder and the format, for example "Ninja · ProRes RAW".

Everything shows in the photo bar and is saved separately in its data.

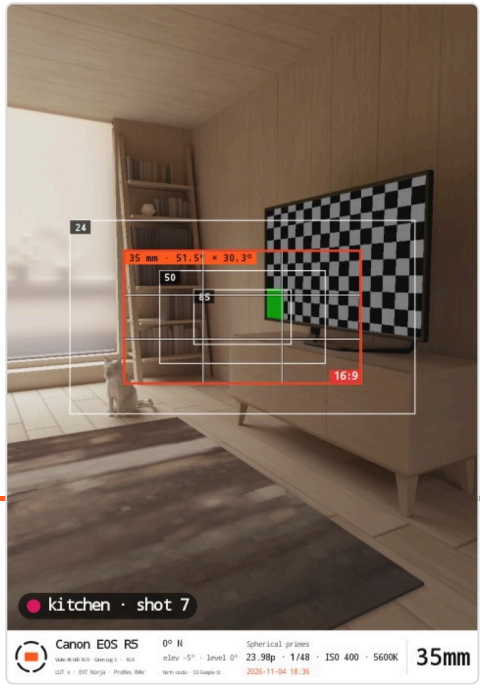
7. The photo: what Mado saves

Every time you tap the shutter, **two images** are saved in `Pictures/Mado`:

- `_marco`**:** the photo with the frames of the whole kit, the guides and the delivery line.
- `_plano`**:** the photo cropped exactly to the current frame, as it would come out.

The **bottom bar** carries: camera and mode (with the colour profile), the phone lens or your note, the heading and tilt, the location name (if a project is selected) or the coordinates, the lens set or the operator settings, the date and time, and the **focal length in large type**. The **scene tag** ("kitchen · shot 7") sits at the bottom left of the image.

If you moved the phone while the photo was being taken, a notice tells you: the photo may come out shifted.



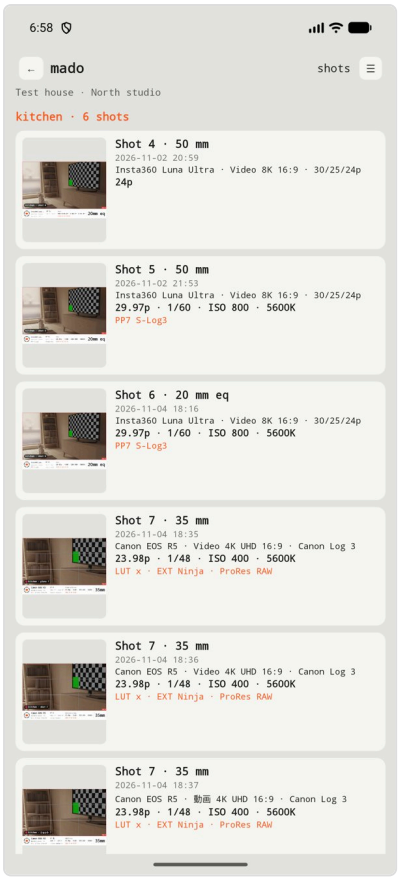
A "marco" photo with its data bar

8. Projects, locations and scenes

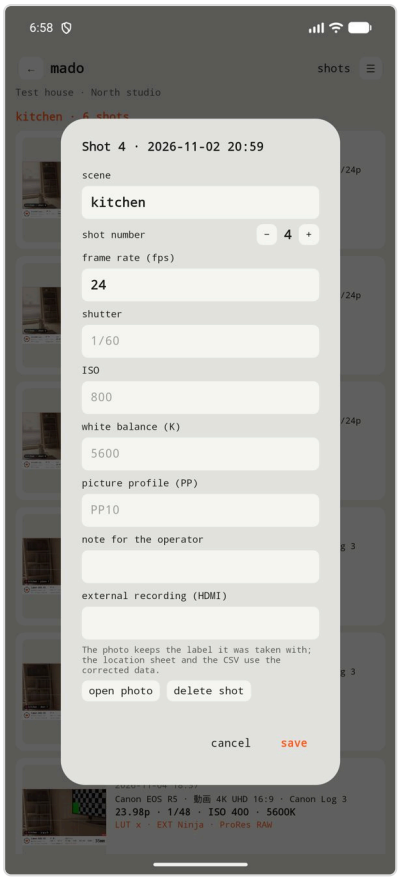
Menu ≡ → **projects and locations**.

1. **Project:** type the name and tap **+**.
2. **Location** (a studio, a house…): type the name and tap **+**. Write the **address** and the **logistics notes** (parking, dressing room, art department…).
3. In the options panel you choose the active project and location. Every photo is tied to them.
4. The **scene** (for example "fireplace") and the **shot number** are typed in the panel. After each photo the number goes up by itself. Scenes already used in the location appear as shortcuts and continue their numbering.

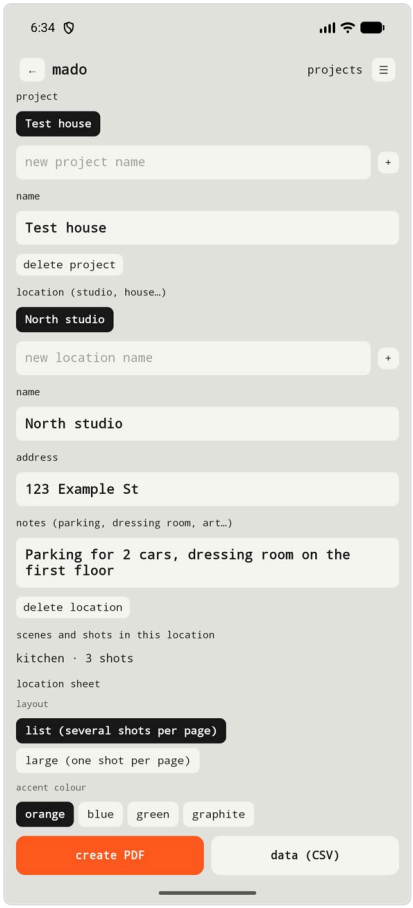
1. **View and edit the shots:** on the same screen, tap **view and edit shots**. You will see each shot with its thumbnail, grouped by scene. Tap one to correct its **scene**, **shot number** and **operator settings**, to **open the photo** or to **delete it**. The photo keeps the label it was taken with; the location sheet and the CSV use the corrected data. When deleting, you choose between removing it only from the sheet (the photos stay in the gallery) or deleting its photos too (this cannot be undone).



The shots of a location



Correcting a shot



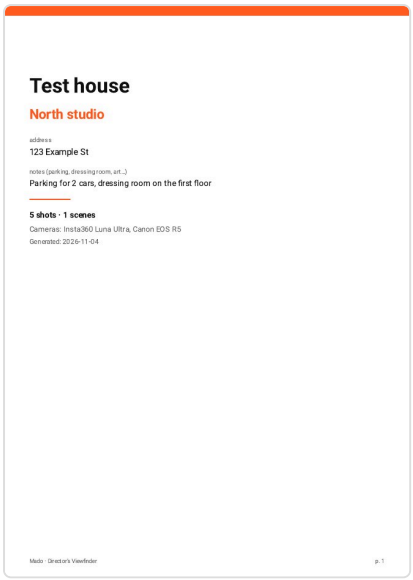
The projects screen

Deleting a project or a location only removes it from the list: **the photos stay in the gallery.**

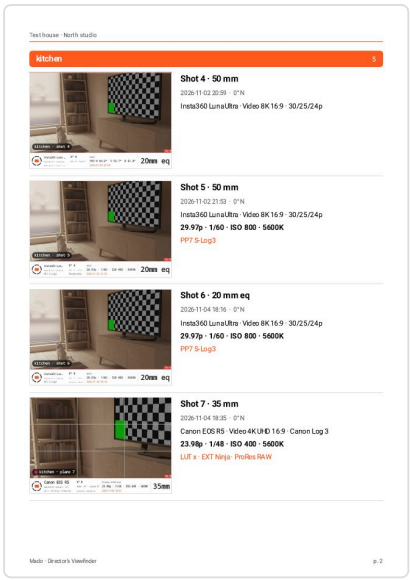
9. Location sheet (PDF and CSV)

On the projects screen, below the scenes, there is the **location sheet** section.

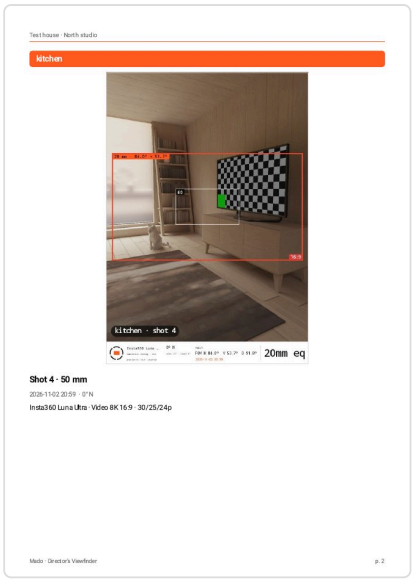
1. Choose the **layout:** *list* (several shots per page) or *large* (one shot per page, with the full frame view).
2. Choose the **accent colour** (orange, blue, green or graphite).
3. Tap **create PDF**. It is saved in `Downloads/Mado` and the share menu opens.



Cover



List layout



Large layout

The PDF is A4 on a white background. The cover carries the project, the location, the address, the notes, the totals and the cameras used. Shots are grouped by scene, with their photo, shot number, focal length, time, heading, camera and mode, the operator settings and the notes. It includes Japanese text. If a photo was deleted from the gallery, the sheet says so in its place.

Data (CSV): one row per shot, for those who prefer to build the document in Keynote, Canva or a spreadsheet.

10. Calibration: why it matters and how to do it

The frame is calculated from the **field of view of the phone lens**. Each phone reports that figure with an error of a few degrees, larger on wide-angle lenses. When a lens is **calibrated** (measured), the frame is accurate. When it is not, the notice appears and the photo records it as "not verified".

If your phone is not calibrated, you have two options from the notice:

- **Share diagnostic:** sends the phone's technical file (model, Android version and data about its cameras, no photos or location) so we can add your model.
- **Calibrate lenses:** you measure the lens yourself (about 10 minutes per lens).

How to measure a lens

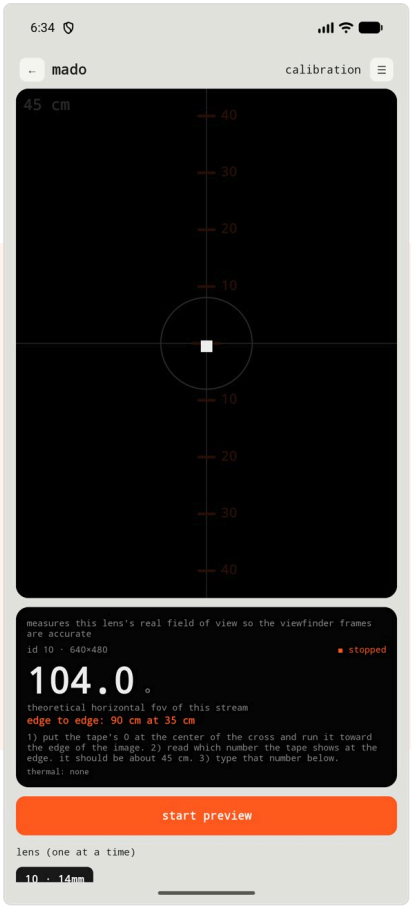
You need a **tape measure**, adhesive tape and a flat wall. Menu ≡ → **calibrate lenses**.

1. Choose the **lens**, one at a time. The app suggests the **distance** to the wall.
2. Stick the tape on the wall horizontally, at the height of the lens. Place the phone (in landscape) at that distance, **measured from the lens glass**, pointing straight at it and resting on something steady.
3. Tap **start preview**. Put the tape's **0 under the cross** in the centre and run it toward one edge of the image.
4. Read **which tape number sits right at the image edge** and type it. Tap **save**.

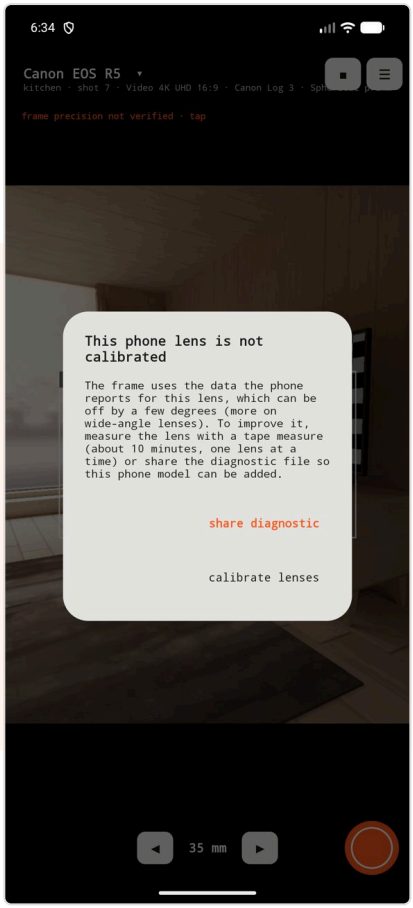
5. Tap **stop** before switching lens and repeat with each one.
6. When you finish, tap **share** to send the readings.

> **Tips.** The orange numbers on the screen are a guide ruler calculated with the theoretical value: the number that matters is the one at the edge. A distance error weighs a lot on wide-angle lenses; measure twice. Telephoto lenses may not focus up close: use 1 m or more. If the phone gets warm, stop.

The screen also shows the phone's **internal calibration** (if it provides one), so you can compare it with what you read on the tape.



The calibration screen



The calibration notice

11. Heat, battery and privacy

- **To avoid heating the phone,** the preview is light and has limits: it pauses after **90 seconds without touching the screen**, stops after **5 minutes** and switches off if the phone gets **hot** (you will see a notice). Resume with **start preview**.
- **The full-resolution photo** is only taken when you tap the shutter.
- **Privacy:** no ads, no tracking and no internet permission. Photos, projects and settings stay on your phone. The diagnostic file is only shared if you send it. The full policy is in `docs/politica-privacidad.md`.

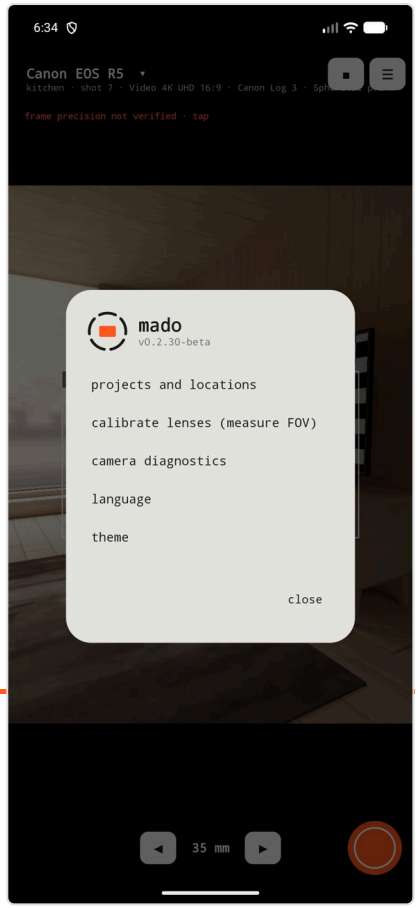
12. Language and theme

Menu ≡ → **language** (automatic, español, English or 日本語) and ≡ → **theme** (automatic, light, dark or **red** for dark sets).

13. Troubleshooting

- **The frame does not match what I see.** Calibrate the lens (chapter 10). Without calibration the error can be a few degrees.

- **The background comes out blurred.** Tap the subject you want in focus on the screen and read the distance that appears next to the marker. The focus stays locked until your next tap. To go back to continuous focus, restart the preview.
- **"The phone moved" notice.** Keep the phone still until " ✓ saved" appears.
- **The camera switched off by itself.** It was inactivity (90 s), the 5-minute limit or heat.
- **A photo says "photo not available" in the PDF.** The original photo is no longer in the gallery.
- **I need a format that is not there.** Use "custom" and type the sensor, and tell us which camera and mode you are missing.



The menu

14. Glossary

- **FOV (field of view):** how much a lens covers, in degrees.
- **Delivery aspect:** the final aspect ratio of the video (for example 16:9).
- **Squeeze factor:** how much an anamorphic lens compresses the image horizontally (2x, 1.33x).
- **Log:** a flat recording curve that leaves more room when grading colour (S-Log3, Canon Log 3, I-Log…).
- **Picture Profile (PP):** an image setting stored in Sony cameras.
- **180° rule:** shutter speed equal to half the time of each frame (1/48 s at 24 fps).
- **Full-frame equivalent:** the focal length that would give the same field on a 36 mm wide sensor.

15. Feedback

Mado is in beta and your experience improves the app. Tell us which cameras you use, what data you need to see and what fails. Write to whoever invited you to the beta. *(Contact email: to be defined before launch.)*